Assignments for the Dynamical Systems Course

May 28, 2026

1 Assignments set II

1.1

Consider one-dimensional population dynamics

$$\dot{x} = rx(1 - \frac{x}{K}) - H$$

with $r, k, H > 0$, $x \geq 0$.
a) Assume $H < \frac{rK}{4}$. Draw phase-portrait on real line. What level of initial population (x_0) never dies out?
b) What happens to $x(t)$ if $H > \frac{rK}{4}$?

1.2

$$\begin{aligned} \dot{x} &= x^2 - y \\ \dot{y} &= x - y \end{aligned}$$

a) Find fixed points.
b) Draw nullclines and illustrate the directions of the flaw in different regions of phase space.
c) Sketch by hand a rough phase portait of the system.
d) Use Matlab/Python to draw enough of trajectories to illustrate the main features of the phase portrait.

1.3

Consider a pair of 'cooperating' populations:

$$\begin{aligned} \dot{x} &= x(K + ay - cx) \\ \dot{y} &= y(L + bx - ey) \end{aligned}$$

a) Show there is a fixed point (x^*, y^* with $x^*, y^* > 0$.
b) Show that this fixed point is stable.
c) Draw nullclines and phase portrait for $x^*, y^* > 0$. Discuss the basin of attraction of the stable fixed point and stable and unstable manifolds for another fixed point in the system.

1.4

$$\begin{aligned} \dot{x} &= -\beta xy \\ \dot{y} &= \beta xy - \gamma y \end{aligned} \tag{1}$$

a) Find fixed points and nullclines.
b) Show that there is a function $G(x, y)$ that is constant along trajectories (hint: consider $\frac{\dot{y}}{\dot{x}}$).
c) Draw the phase portrait for $x, y > 0$ using $G(x, y)$

1.5

$$\begin{aligned} \dot{x} &= y \\ \dot{y} &= -x + x^3 \end{aligned}$$

a) Find fixed points
b) Find potential energy $V(x)$ such that $\dot{y} = -\frac{\partial V}{\partial x}$ and draw its graph. Use physics intuition to classify each fixed point.
c) Show that energy $E(x, y) = \frac{y^2}{2} + V(x)$ is preserved on each trajectory. Use this to draw phase portrait of the system.

1.6

$$\begin{aligned} \dot{x} &= -x^3 + xy^2 \\ \dot{y} &= -2x^2y - y^3 \end{aligned}$$

a) Show that the function $L(x, y) = x^2 + y^2$ is the Lyapunov function for the region $x, y \neq 0$ (i.e. $\dot{L} < 0$ along each trajectory).
b) What does this tell about solutions of the system?